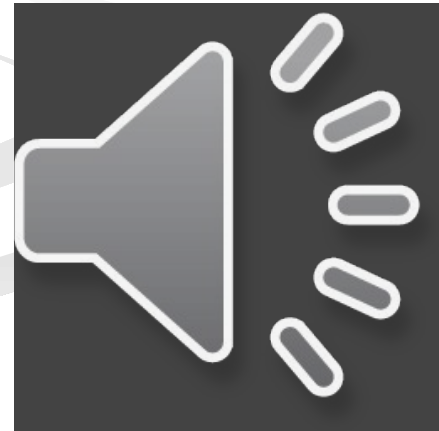


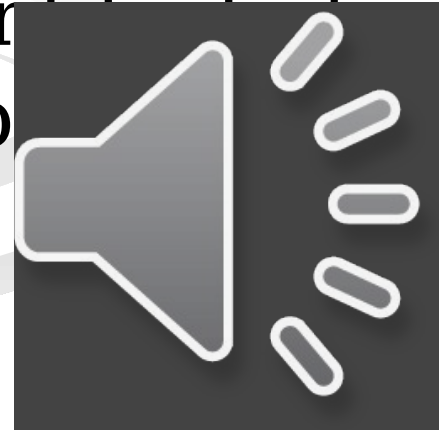
The meta-analysis of dependent effect sizes: Using RVE

**Master of Statistics
KU Leuven**

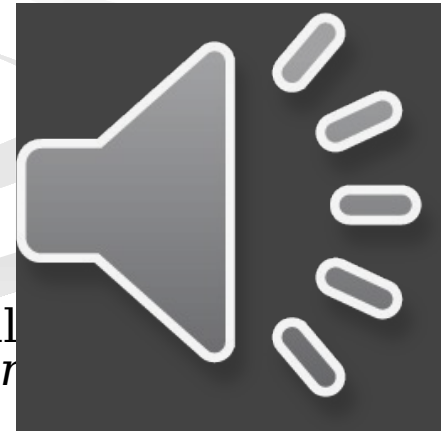
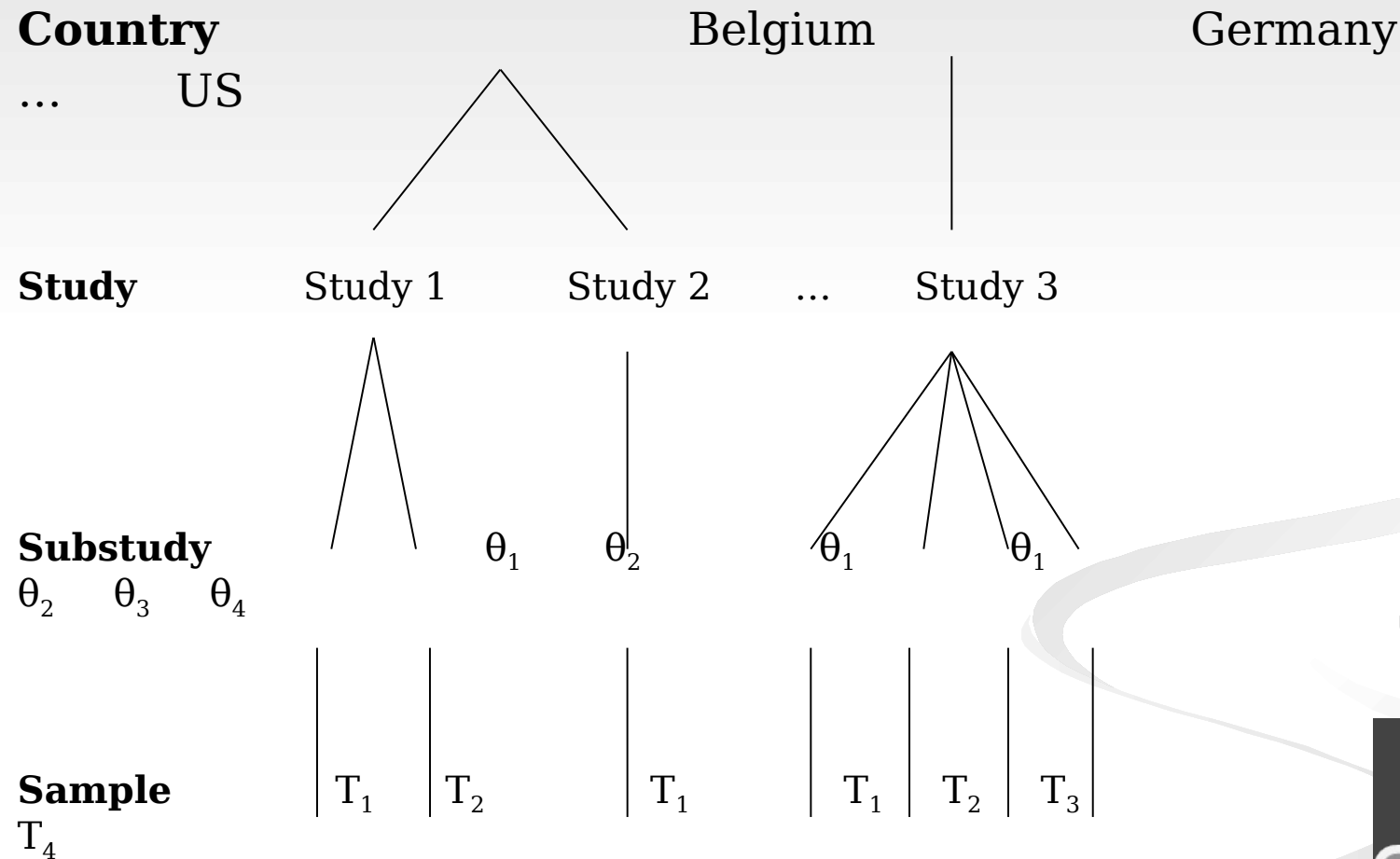


Recap

- Traditional MA models assume independent effect sizes
- Dependence across studies (e.g., multiple studies from same research group)
- Dependence within studies, e.g.:
 - Multiple time points
 - Multiple outcome variables
 - Multiple variations of treatment
- Multilevel modelling can be helpful



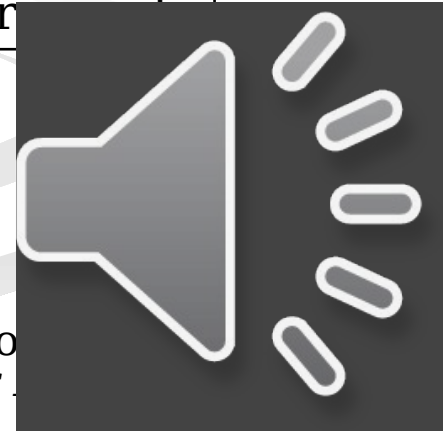
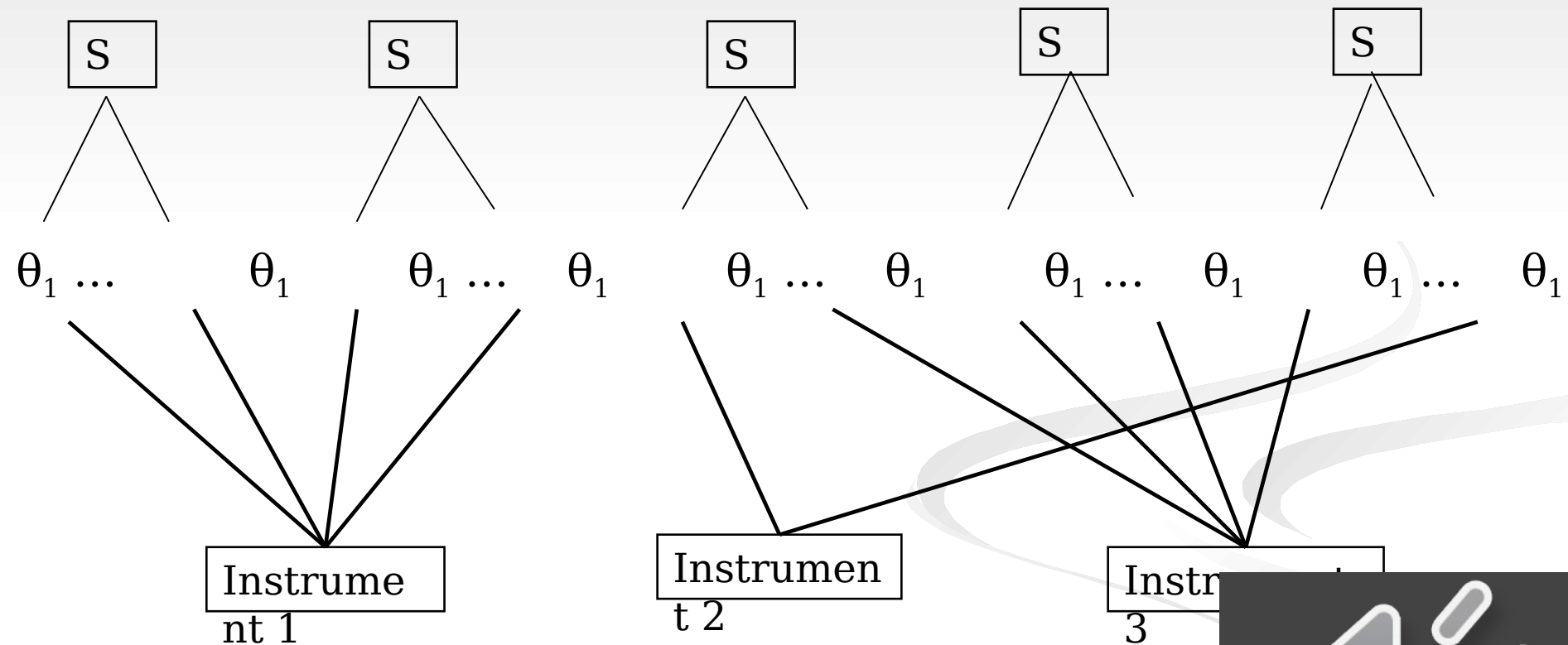
Multilevel meta-analyses ...



Example:

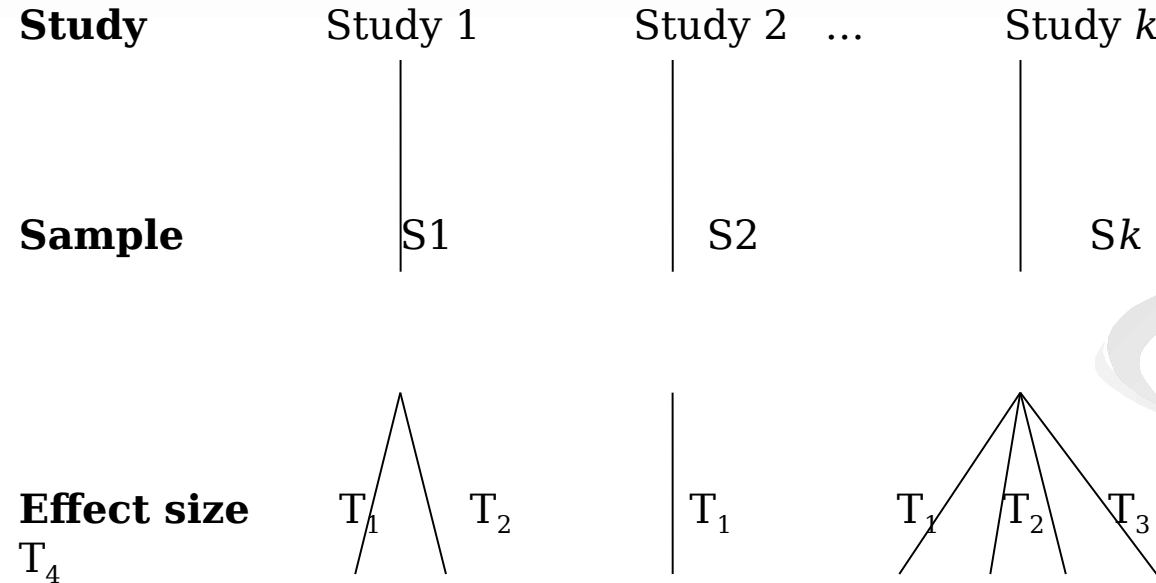
Kende, J., et al. (2018). Equality revisited: A cultural meta-analysis of intergroup contact and prejudice. *Social Psychological and Personality Science*, 9, 887-895.

Non-hierarchical multilevel meta-analyses ...



Fernández-Castilla, B., et al. (2019). A demonstration and evaluation of cross-classified random effects models for meta-analysis. *Behavior Research Methods*, 51, 1286-1304.

Dependencies due to multiple outcomes per sample



⇒ Sampling covariance

Multiple SMDs within one study for different outcomes:

$$\sigma_{g_j}^2 \cong \frac{n_E + n_C}{n_E n_C} + \frac{\delta_j^2}{2(n_E + n_C)} \quad \text{unknown}$$

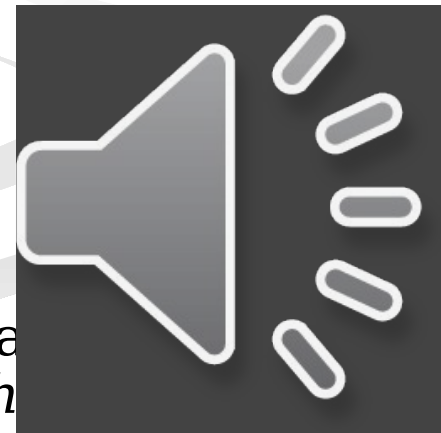
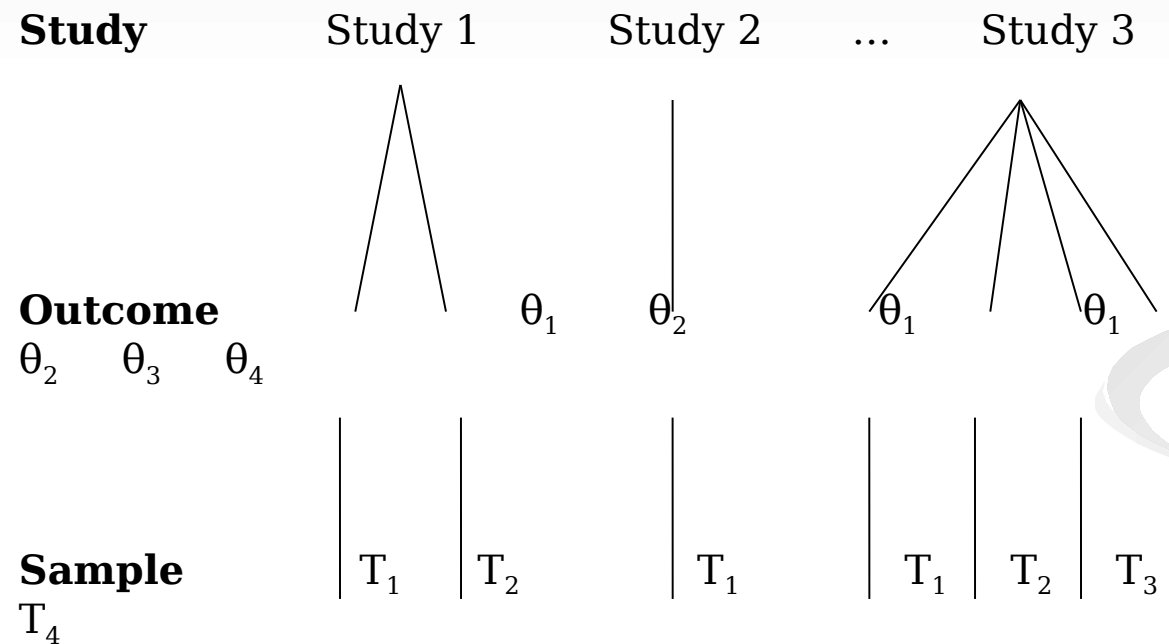
$$\sigma_{g_j g_{j'}} \cong \frac{n_E + n_C}{n_E n_C} \rho_{jj'} + \frac{\delta_j \delta_{j'} \rho_{jj'}^2}{2(n_E + n_C)}$$

⇒ Multivariate meta-analysis

- Using a reasonable guess of ρ (e.g., 0.5)
- Using sensitivity analyses



Alternative: assume this structure



Van den Noortgate, W., Lopez-Lopez, J. A., Marin-Ma
 Sánchez-Meca, J. (2013 & 2015). *Behavior Research*

But ...

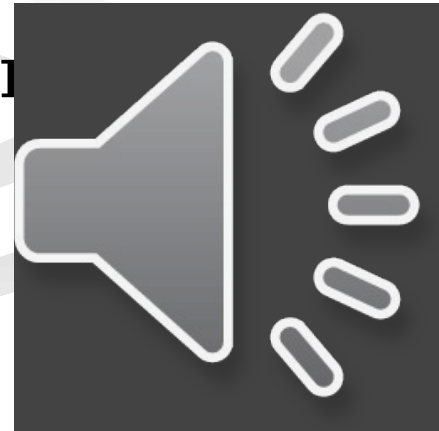
- Multilevel model makes assumptions:
 - Correct specification of the data structure
 - For multiple effect sizes per sample, we consider effects as coming from independent samples!
 - Sufficient sample sizes at each level
 - Independence at each level
 - Normality at each level



In addition: possible variance issues

E.g., for pre-post studies, correlation between pre- and post-measurements are not reported

- ⇒ Sampling variance of observed effect sizes may be invalid
- ⇒ Effect sizes may be inappropriately weighted
- ⇒ Standard errors, and hence inference model parameters may be wrong



Robust Variance Estimation (RVE) as a safety net!

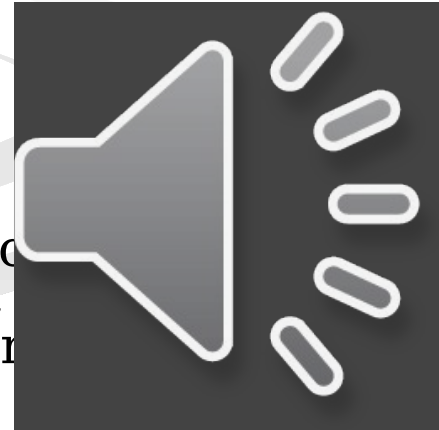
- Instead of using **model-based** standard errors, let us use the **observed** variation in effect sizes to estimate the standard errors of the meta-analytic results.
- Makes standard errors robust for model-misspecification
- Also known as:
 - Empirical standard errors
 - Huber-White standard errors
 - Cluster robust variance estimation



RVE

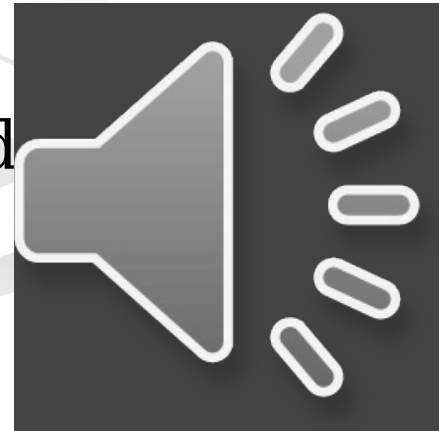
- Observed residuals: $e_{ij} = T_{ij} - \theta$
- Instead of assuming $\sigma_T^2 = \sigma_{between}^2 + \sigma_{within}^2 + \sigma_{T_i|\theta_i}^2$,
we estimate:
 - $\hat{\sigma}_T^2 = s_e^2$
 - $\hat{\sigma}_{e_{ij}e_{i'j}} = s_{e_{ij}e_{i'j}}$
 - $\hat{\sigma}_{e_{ij}e_{ij'}} = 0$

Hedges, L.V., Tipton, E., and Johnson, M.C. (2010) Robust variance estimation in meta-regression with dependent effect sizes. *Research Synthesis Methods*, 1(1), 39-65. (with an erratum 164-165.)



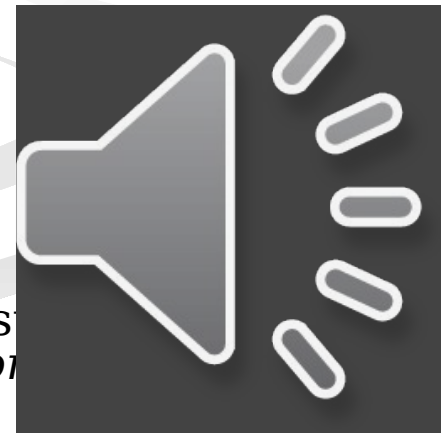
Value

- RVE does lead to appropriate standard errors, even if
 - the covariance structure (independence within hierarchical levels) is mis-specified,
 - the normality assumption is violated, or
 - the weights are not inverse-variance.
 - RVE does not lead to improved
 - estimates of the variances
 - weights
- ⇒ Combine RVE with a reasonably good



“Working” models (Pustejovsky & Tipton,)

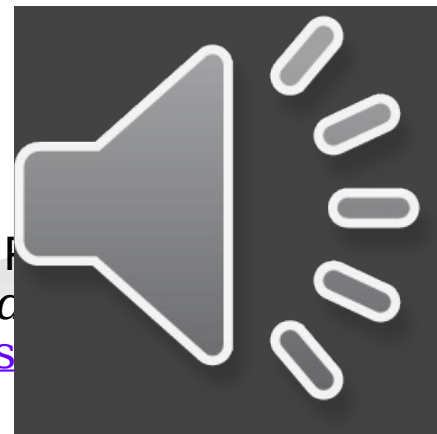
- Hierarchical effects (HE) model
- Correlated effects (CE) model
- Hierarchical and correlated effects (CHE) model



An example

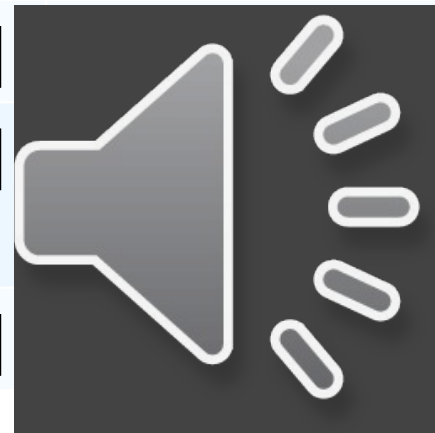
- Correlation between de-humanization and violence studied in MA of Landry et al. (2024)
- Projects – studies – samples – effects studied – sampling
 - 5-level model?
 - 4-level model accounting for sampling covariance?

Landry, A.P., Ghezae, I., Abou-Ismaïl, R., Spooner, S., Auguste, R., Ragnhildstveit, A., Van den Noortgate, W., Gelfand, M.J., & Seli, E. (in press, publication). *The uniquely powerful impact of explicit, blatant dehumanization on support for intergroup violence*. Preprint available at https://osf.io/rxiv/weh7k_v1.



An example

Model	Overall effect	95% CI	CI Width
2-level	.321	[.305, .337]	.032
+ RVE		[.296, .346]	.050
5-level	.345	[.311, .379]	.068
+ RVE		[.310, .380]	.069
CHE (rho = .5)	.343	[.309, .376]	.067
+ RVE		[.309, .377]	
CHE (rho=.8)	.341	[.308, .373]	
+ RVE		[.307, .374]	



Appropriate model + RVE

- Type I error rate under control
- Information on various sources of variance
- Studies appropriately weighted
- ☞ Complexity

